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Research paper

Design of the MDFF-EPA photovoltaic ultra-short-term power prediction algorithm based on FY-4A

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ABSTRACT

The variability of solar radiation significantly impacts the energy output of photovoltaic (PV) power plants, with cloud cover being the primary cause of such fluctuations. To address the issues of power output variability caused by cloud cover and the complexity of ultra-short-term PV power forecasting models, this paper proposes an ultra-short-term PV power forecasting framework and model based on Fengyun-4A (FY-4A) data, named Convolutional-Stacked-LSTM (CS-LSTM). The core task of this framework is first to extract radiation data for specific coordinates, then analyze the corresponding dynamic changes using the Farnebäck optical flow method to capture the movement trends of radiation, forming optical flow input features. Subsequently, this study integrates FY-4A data with optical flow input features and other information to form a comprehensive multimodal input for the CS-LSTM algorithm to predict. The innovation of this research lies in combining multimodal inputs and optical flow features, effectively enhancing prediction accuracy. Additionally, this paper proposes an ultra-short-term forecasting algorithm that combines multidimensional feature fusion with error gradual approximation (MDFF-EPA). By incorporating historical prediction errors into the feature set, the model can progressively correct and optimize prediction results, improving prediction accuracy and stability. Experimental validation using real operation data from the Huadian Hubei Suixian Yindian PV power plant demonstrates that, compared to the control group, the experimental group with optical flow input features achieved over a 1% improvement in prediction accuracy for certain months. Moreover, in the comparative analysis of prediction accuracy across four weather types – clear, cloudy, rainy, and heavy rain – the experimental group showed a reduction in RMSE (Root Mean Square Error) percentage by 15.04%, 11.97%, 14.84%, and 18.86% respectively. Future research could further consider capturing cloud movement information and improving the optical flow model to enhance the accuracy of ultra-short-term PV power forecasting.

1. Introduction

With the global shift in energy structures and the demand for environmental protection, photovoltaic (PV) power generation, as an energy utilization mechanism that fully leverages solar energy, holds particularly promising prospects in the energy market (Luo et al., 2021). However, PV power generation is affected by various factors such as weather conditions and seasonal changes, exhibiting irregular periodicity and volatility (Li et al., 2022), which increases the difficulty of grid system scheduling and management. Therefore, accurate ultra-short-term PV power forecasting is crucial for the stable operation of power systems and the effective functioning of the power market. Ultra-short-term power forecasting can provide precise prediction information for grid dispatching, helping grid operators better manage the power system, reduce operational costs, and improve the stability and reliability of the power system.

In recent years, the installed capacity of PV power systems worldwide has grown rapidly. From 2005 to 2018, PV installed capacity increased 100-fold (Maka and Alabid, 2022), and by 2020, the global installed PV power capacity exceeded 700 GW (Rozon et al., 2023), while the installation cost of PV systems decreased by over 81.2% from 2010 to 2020 (Silva et al., 2023). This trend not only drives the advancement of PV power generation technology but also increases the demand for high-accuracy PV power forecasting.

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The mainstream methods for ultra-short-term PV power forecasting include physical methods (Ogliari et al., 2017), statistical methods (Ni et al., 2019), and machine learning methods (De Giorgi et al., 2015). Physical methods are based on the physical explanation of the working principles of PV cells and meteorological conditions, combined with meteorological data to establish physical models and equations for power forecasting. However, due to high modeling costs and computational complexity (Huang et al., 2010), there are many difficulties in the application of physical methods in ultra-short-term PV power forecasting research. Traditional statistical forecasting methods involve fitting and correlation analysis of historical data such as PV output to establish a correlation mapping relationship between data and models (Wang et al., 2019) for PV power forecasting. Methods such as Autoregressive Moving Average Model (ARMA) (Jiang et al., 2021), Autoregressive Integrated Moving Average Model (ARIMA) (Zhou et al., 2017), and Fuzzy Theory (Halabi et al., 2018) are used. However, due to the significant influence of climate and other factors on PV power generation, statistical analysis models tend to have large deviations when PV power changes greatly or data is unstable (Liu et al., 2020).

In recent years, with the development of artificial intelligence and big data technology, machine learning methods have been widely applied in ultra-short-term PV power forecasting (Sobri et al., 2018), such as Support Vector Machines (SVM) (Pan et al., 2020), Artificial Neural Networks (ANN) (Voyant et al., 2017), Extreme Learning Machines (ELM) (Jing et al., 2017), and Extreme Gradient Boosting (XGBoost) (Zheng and Wu, 2019). These methods have strong nonlinear mapping capabilities and can handle complex nonlinear relationships and complex data structures, improving prediction accuracy (Nespoli et al., 2019). Among machine learning algorithms, deep learning algorithms, which have the ability to extract deep features, can better understand the change trends of time-series data (Liu et al., 2021), further enhancing prediction accuracy.

For short-term PV power forecasting, methods relying on historical power generation and numerical weather predictions can meet the needs. However, for ultra-short-term PV power forecasting, it is necessary to further consider the factors of irregular cloud movement and include the shading conditions of clouds on PV power plants in the prediction models. Incorporating related input factors that consider cloud movement into prediction models can improve the accuracy of ultra-short-term PV power forecasting (Gandoman et al., 2018). For instance, literature (Wang et al., 2024a) proposed an ultra-short-term PV power forecasting method based on ground cloud images and chaos characteristics, which improves prediction accuracy under abrupt radiation scenarios by introducing a two-stage cloud speed calculation model and a dynamic modeling method based on chaos theory. Literature (Bo et al., 2023) proposed an ultra-short-term PV power forecasting method based on multi-exposure high-resolution sky images and deep learning, which improves prediction accuracy and stability through image fusion, sliding window cutting and stitching strategy, and Convolutional Long Short-Term Memory (CLSTM) model. Literature (Limouni et al., 2023) proposed an Long Short-Term Memory and Temporal Convolutional Network (LSTM-TCN) model for one-step and multi-step PV power forecasting, showing that this model outperforms using Long Short-Term Memory Networks (LSTM) and Temporal Convolutional Network (TCN) models alone under various seasons and weather conditions. Literature (Wang et al., 2024b) proposed an ultra-short-term PV power forecasting method based on the Variational Mode Decomposition-Complete Ensemble Empirical Mode Decomposition with Adaptive Noise-Long Short-Term Memory (VMD-CEEMDAN-LSTM) model, enhancing the capability to handle complex datasets through advanced multi-modal decomposition and achieving high prediction accuracy. Literature (Huang et al., 2022) developed a hybrid deep learning-based multi-feature ultra-short-term PV power forecasting model, combining the advantages of Convolutional Neural Networks (CNN) and LSTM. Literature (Huang and Yang, 2023) used a Memory Long Short-Term Time Series Network (MLSTM) for ultra-short-term PV power forecasting, showing significant improvement in prediction performance. Additionally, literature (Zang et al., 2024) proposed a novel data-driven ultra-short-term PV power forecasting method, combining SVM and CNN, and introducing a dynamic updating mechanism to improve the real-time adaptability of the prediction model, significantly enhancing prediction accuracy. Literature (Jiang et al., 2024) combined the LSTM with the Dynamic Time Warping (DTW) method, improving the accuracy and robustness of PV power forecasting, especially under irregular weather conditions. Literature (Zhang et al., 2023) proposed an ultra-short-term PV power forecasting model based on the CEEMDAN (Complete Ensemble Empirical Mode Decomposition with Adaptive Noise), FE (Fuzzy Entropy), FIG (Fuzzy Information Granulation), ARIMA (Autoregressive Integrated Moving Average), and WOA (Whale Optimization Algorithm)-optimized LSTM (Long Short-Term Memory) hybrid model. This model enhances prediction accuracy by aggregating the outputs of various components, thereby precisely capturing the actual PV power values. However, the model is encumbered by an intricate data preprocessing workflow, complex parameter configurations, substantial computational overhead, and diminished interpretability due to the integration of multiple sophisticated algorithms. Literature (Wang and Lin, 2023) proposed an ultra-short-term PV power forecasting model using VMD (Variational Mode Decomposition) and ELM key parameters optimized by ISSA (Improved Salp Swarm Algorithm), effectively mining historical data, but only considered single meteorological factors without considering the impact of weather conditions like cloud movement on the prediction.

Despite certain progress made by existing ultra-short-term PV power forecasting methods, there are still many limitations in dealing with complex meteorological conditions and dynamically changing solar irradiance. Physical and statistical methods show deficiencies in handling nonlinear and non-stationary data, and traditional machine learning methods also face limitations when dealing with multimodal and high-dimensional data. Moreover, these methods' prediction performance significantly decreases under rapidly changing weather conditions. Therefore, this paper proposes an innovative ultra-short-term forecasting algorithm combining Multidimensional Feature Fusion and Error Progressive Approximation (MDFF-EPA). The core innovation of MDFF-EPA lies in combining multidimensional feature fusion and a dynamic error feedback mechanism, incorporating historical prediction errors into the feature set, allowing the model to more accurately capture and correct biases in predictions, thereby gradually improving prediction accuracy. This innovative error feedback mechanism enhances not only the accuracy of single predictions but also the stability of continuous predictions by the model.

As an application example of this algorithm, this paper also proposes a model based on FY-4 A satellite data, Convolutional-Stacked-LSTM (CS-LSTM). By combining the advantages of Convolutional Neural Networks (CNN) and Long Short-Term Memory Networks (LSTM), CS-LSTM effectively captures spatial and temporal features, thereby improving prediction accuracy. This paper will describe in detail the proposed ultra-short-term PV power forecasting framework based on FY-4 A and MDFF-EPA, and demonstrate its superior performance under various weather conditions through experiments.

The rest of this paper is organized as follows. Section 2 introduces and analyzes the relevant features and capabilities of FY-4 satellite and the Farnebäck optical flow algorithm. Section 3 presents the ultra-short-term forecasting algorithm combining Multidimensional Feature Fusion and Error Progressive Approximation, as well as the constructed ultra-short-term PV power forecasting framework based on FY-4 A. Section 4 evaluates the effectiveness and applicability of optical flow input factors and the prediction performance of the proposed model under all-weather conditions, providing experimental results. Finally, Section 5 summarizes this work and points out areas for future improvement.

2. Related work

2.1. FY-4 satellite data

The FY-4 series satellites are a new generation of geostationary orbit meteorological satellites independently developed by China. The primary data processing mainly includes data preprocessing, calibration, and inversion steps. The first satellite, FY-4 A, carries four advanced monitoring instruments: Advanced Geosynchronous Radiation Imager (AGRI), Geostationary Interferometric Infrared Sounder (GIIRS), Lightning Mapping Imager (LMI), and Space Environment Monitor (SEP) (Yang et al., 2017), covering different wavelength ranges comprehensively. Compared to the Fengyun-2 (FY-2) satellite, the addition of 9 spectral bands and high spatial and temporal resolution (Yang et al., 2017) have improved the monitoring accuracy and resolution of solar radiation. This provides scientific data support for the research in this paper. For subsequent satellites in the FY-4 series, the number of AGRI channels has increased from 14 to 18, the infrared spatial resolution has been improved to 2 kilometers, and the frequency of acquiring full-disk satellite cloud images has been increased from 15 min to 5 min (Yang et al., 2017), significantly enhancing the monitoring capability for solar radiation.

2.2. Farnebäck optical flow algorithm

The Farnebäck optical flow method is a traditional computer vision technique that utilizes the concept of dense optical flow, employing changes in pixel intensity between adjacent frames to compute the flow of light. The method presupposes that image gradients are constant and local optical flow is steady (Choi et al., 2022).

If the input image is viewed as a function of two variables, then:

$$f(\mathrm{x}) = \mathrm{x}^{\mathrm{T}}\mathrm{A}\mathrm{x} + \mathrm{b}^{\mathrm{T}}\mathrm{x} + \mathrm{c} \tag{1}$$

where $f(x)$ represents the neighborhood information matrix, X is the two-dimensional coordinate of the cloud image, A is a 2×2 symmetric matrix, b is a 2×1 matrix, and A, b, and c are the relevant coefficients of the quadratic polynomial matrix function.

If the original image is:

$$f_1(\mathrm{x}) = \mathrm{x}^{\mathrm{T}}\mathrm{A}_1\mathrm{x}\mathrm{b}_1^{\mathrm{T}} \tag{2}$$

Then the image after pixel movement can be represented as:

$$\begin{aligned} f_2(x) &= f_1(x-d) \\ &= (x-d)^T A_1 (x-d) + b_1^T (x-d) + c_1 \\ &= x^T A_2 x + b_2^T x + c_2 \end{aligned} \tag{3}$$

Where d is the assumed positional offset between consecutive image frames, and A_1, A_2, b_1, b_2, c_1, and c_2 are the relevant coefficients of the quadratic polynomial matrix function.

According to the optical flow method, the following assumptions are made:

$$\begin{cases} A_2 = A_1 \\ b_2 = b_1 - 2A_1 d \\ c_2 = d^T A_1 d - b_1^T d + c_1 \end{cases} \tag{4}$$

Then, the displacement d can be obtained as:

$$d = -\frac{1}{2} A_1^{-1} \left(b_2 - b_1\right) \tag{5}$$

In PV power prediction, the Farnebäck optical flow algorithm can be used to estimate cloud movement (Teerakawanich et al., 2020), thereby predicting future solar radiation. This is because the movement of clouds affects the reception of solar radiation, and solar radiation is a key factor influencing PV power generation (Wang et al., 2018) .

2.3. Hybrid CNN-LSTM model

Currently, hybrid CNN-LSTM (Convolutional Neural Network and Long Short-Term Memory) structures are used for time series prediction in various applications across different fields, such as the solar energy prediction domain (Gao et al., 2020) . CNNs can effectively extract spatial features and identify patterns in images, while LSTMs can capture long-term dependencies and dynamic changes in time series data. However, in the field of PV power prediction, due to the high demand for computational resources and lower accuracy of traditional CNN models, hybrid models of CNN and LSTM have emerged as a solution to these problems, with many researchers proposing various combinations of CNN-LSTM (Elizabeth Michael et al., 2022) . Ref. Elizabeth Michael et al. (2022) proposed a multi-step CNN-LSTM model to predict short-term solar power, achieving better prediction results in the prediction of solar irradiance and Plane of Array (POA) compared to existing models based on deep learning techniques. Zhu et al. (2021) proposed a SCNN-LSTM model to predict Direct Normal Irradiance (DNI) within 10 min, with better forecasting performance on sunny and rainy days than some existing methods.

3. Model proposal

Cloud movement is a dynamic, nonlinear phenomenon that can be processed using artificial neural networks. To predict the ultra-short-term power of photovoltaics at a specific location, this paper utilizes deep learning and optical flow techniques to design a data-intensive framework matched to FY-4 A data. The core innovations of the model proposed in this paper are two-fold: first, it introduces an ultra-short-term prediction algorithm that combines multi-dimensional feature fusion with error gradual approximation; second, it designs a data-intensive framework matched to FY-4 A data, using radiation motion information extracted from FY-4 A data as input features and validating with real PV output data.

3.1. Ultra-short-term prediction algorithm combining multi-dimensional feature fusion with error gradual approximation

The core innovation of the Multi-Dimensional Feature Fusion and Error Gradual Approximation (MDFF-EPA) algorithm lies in combining multi-dimensional feature fusion with a dynamic error feedback mechanism. It incorporates historical prediction errors into the feature set, allowing the model to more accurately capture and correct biases in predictions, thereby gradually improving prediction accuracy. Specifically, the model first calculates the deviation between predicted and actual values in historical data, then uses this error information as input features, combined with other meteorological and photovoltaic data for the next stage of prediction. This method allows each round of prediction to be corrected and optimized based on the previous round, effectively reducing long-term cumulative errors and improving the overall accuracy and stability of ultra-short-term power prediction. This innovative error feedback mechanism not only enhances the accuracy of single predictions but also improves the stability of the model's continuous predictions. The specific method is as follows:

(1) Error Calculation and Fusion

First, calculate the error between the model's predicted values and the actual PV power output for each prediction period, then fuse these error values with other features to form a comprehensive feature set;

(2) Model Iterative Optimization

In the next prediction cycle, the prediction error from the previous round is used as a new input feature to retrain the model. This method allows the model to self-correct in each iteration, gradually reducing the prediction error.

(3) Dynamic Adjustment Strategy

By using an adaptive learning rate and optimization algorithms, as well as an early stopping training method, the weight of historical errors in the model is dynamically adjusted, ensuring that the model maintains high predictive performance under different conditions.

The Farnebäck optical flow method can not only estimate the motion displacement of pixels but also provide information on the motion speed and direction of pixel points. The formula is as follows:

$$\begin{cases} V = \sqrt{u^2 + v^2} \\ \theta = \arctan\left(\frac{v}{u}\right) \end{cases} \tag{6}$$

where u and v represent the displacement in the horizontal and vertical directions, respectively. By calculating the motion speed V and θ the change in direction for each pixel, more detailed dynamic features can be obtained, which is especially important for capturing the movement characteristics of clouds. In this paper, these dynamic features are integrated into the model, which can greatly enrich the feature set and improve the accuracy of ultra-short-term prediction.

In the feature fusion section, this paper defines the feature vector X_t as a combination of observed data at time step t, NWP (Numerical Weather Prediction) data, and various features extracted from FY-4 A satellite radiation images. Therefore, feature fusion can be represented as:

$$X_t = \mathrm{CNN}(\mathrm{Satellite}) \oplus \mathrm{LSTM}(\mathrm{NWP}) \tag{7}$$

where $\oplus$ represents the feature concatenation operation.

Assuming P_t is the actual PV power output at time t, and $\hat{P}_t$ is the model's predicted power input, then the prediction error is:

$$e_t = P_t - \hat{P}_t \tag{8}$$

Take e_t as input for the next step of prediction, that is:

$$\hat{P}_{t+1} = f(X_t, e_t) \tag{9}$$

where f represents the CS-LSTM model, used for rolling prediction of the next period's PV ultra-short-term power output.

3.2. PV ultra-short-term power prediction framework

Based on FY-4 A data and incorporating the ultra-short-term prediction algorithm combining multi-dimensional feature fusion with error gradual approximation, this paper proposes a method for PV ultra-short-term power prediction. The overall technical scheme is shown in Fig. 1.

As shown in the figure, the flowchart of the PV ultra-short-term power prediction based on FY-4 A involves the following steps:

Step 1 Input:

Multimodal Meteorological Data Collection.

$$W_i = \{\hat{\omega}_1, \ldots, \hat{\omega}_n\} = \{\mathrm{NWP}, \mathrm{FY\text{-}4A}, \mathrm{AP}\} \tag{10}$$

Where NWP, $FY-4A$, and $AP(Actual Power)$ refer to Numerical Weather Prediction data, FY-4 A data, and observed data, respectively.

Step 2 Process:

(1) Acquire three types of radiation data

Select the forecast location, obtain the total radiation, diffuse radiation, and direct radiation data for the forecast position from the FY-4 A data based on the latitude and longitude of the forecast location, and prepare for quality control and fusion;

(2) Radiation movement direction and change speed

Use the Farnebäck optical flow method to process the satellite data with a size of 256 × 256 extracted based on the central latitude and longitude of the forecast location, thereby obtaining the radiation change direction and change speed of the central point and its eight neighboring areas;

Assuming that the motion between pixels can be approximated by a polynomial expansion, the motion is estimated by minimizing the pixel intensity differences between adjacent frames. The core formula can be represented as:

$$E(u, v) = \sum_{x,y} [I(x+u, y+v, t+1) - I(x, y, t)]^2 \tag{11}$$

where $I(x, y, t)$ is the pixel intensity at time t and position(x, y), and u and v are the displacements of the pixel in the horizontal and vertical directions, respectively.

(3) Unify Time Resolution

Use interpolation methods to process the FY-4 A data, generating multiple sets of data, so that their time resolution is unified with that of numerical forecast and observed data;

(4) Multimodal Data Fusion

Fuse FY-4 A radiation data, numerical forecast series data, radiation change direction and change speed data, and observed data to generate the final multimodal input factors;

(5) Ultra-Short-Term Power Prediction

Use the CS-LSTM algorithm for PV ultra-short-term power prediction.

Step 3 Output:

PV power for the next four hours.

As illustrated in Fig. 2, the CS-LSTM, by integrating multiple convolutional layers with multiple LSTM layers, is capable of more effectively processing and analyzing multi-dimensional data in photovoltaic ultra-short-term power prediction. Herein, the convolutional layers enhance the model's capacity in handling spatial features, especially in extracting spatial characteristics from FY-4 A data, while the LSTM layers concentrate on capturing the temporal dependencies within the data, thereby improving the accuracy of temporal predictions. Compared to a singular LSTM approach, the CS-LSTM is more proficient in capturing spatial information such as satellite imagery within the context of photovoltaic ultra-short-term power forecasting.

However, CS-LSTM can only roughly predict the overall motion trend, while dense optical flow methods, such as the Farnebäck optical flow method, can predict a large number of instantaneous motions in small areas of clouds, capturing more motion information. Therefore, incorporating pixel-level motion information into the overall PV ultra-short-term power prediction framework can improve the final accuracy of PV ultra-short-term prediction.

3.3. Evaluation method

To evaluate the accuracy of the prediction method proposed in this paper, the Normalized Root Mean Square Error (NRMSE) is used to verify the accuracy of the prediction. The formula is as follows:

$$\mathrm{NRMSE} = \frac{1}{\mathrm{Cap}} \sqrt{\frac{1}{N} \sum (p_f^i - p_o^i)^2} \tag{12}$$

$$PA = (1 - \mathrm{NRMSE}) \times 100\% \tag{13}$$

where P_f represents the predicted PV power values, P_o represents the actual PV power values, Cap represents the installed capacity, and PA represents the prediction accuracy.

In addition, in the subsequent sections of this paper, the Root Mean Square Error (RMSE) will also be used to measure the degree of fit between the predicted values and the actual observed values. The specific formula is as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N} \sum (p_f^i - p_o^i)^2} \tag{14}$$

$$\text{RMSE Percentage} = \left(\frac{\mathrm{RMSE}}{\overline{P}_{\mathrm{ac}}}\right) \times 100\% \tag{15}$$

where $\overline{P}_{\mathrm{ac}}$ represents the average value of the actual PV output.

To validate the effectiveness and applicability of the proposed method, this paper conducts a comparative analysis using four algorithms: CS-LSTM, LSTM, AE-LSTM (Autoencoder-Long Short-Term

Fig. 1. PV ultra-short-term power prediction method based on FY-4A.

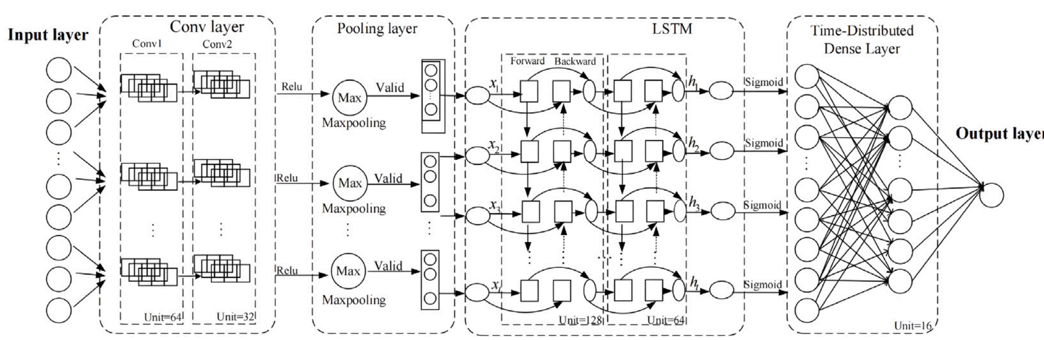

Fig. 2. CS-LSTM network structure.

Memory), and GRU (Gat-ed Recurrent Unit). CS-LSTM(O), LSTM(O), AE-LSTM(O), and GRU(O) represent the control group without the addition of optical flow input features, while CS-LSTM(F), LSTM(F), AE-LSTM(F), and GRU(F) represent the experimental group with the addition of optical flow input features. By comparing the prediction performance of these algorithms under different weather conditions, we can determine the superiority and applicability of the proposed method. The detailed comparative results between the control and experimental groups will be elaborated in subsequent sections.

4. Experiments and discussion

4.1. Experimental procedure

To validate the effectiveness of the proposed ultra-short-term PV power forecasting model based on FY-4 A data, this study selected the Huadian Hubei Suizhou Suixian Yindian PV power plant as the experimental site. This location is in central China, characterized by abundant sunlight, making it suitable for PV power generation. The experiments were primarily conducted on a device equipped with an Intel i5-13490F processor and an NVIDIA 3060ti graphics card with 8 GB of GDDR6 VRAM, running on Windows 11 Professional.

The Huadian Hubei Suizhou Suixian Yindian PV power plant is a large-scale PV power station with an installed capacity of 100MWP. Fig. 3 shows an image of the power plant. The average horizontal total irradiance in Hubei Province in 2023 was 1161.9 kWh/m^2, while in Suizhou City, it was 1353.12 kWh/m^2. Although specific data for Suixian is not available, this information provides an effective reference for this study.

During the modeling process, we utilized three types of radiation data provided by the FY-4 A satellite: total radiation, diffuse radiation, and direct radiation. These data were extracted based on the specific coordinates of Suixian Yindian Town. By integrating these radiation data with other meteorological data, we conducted detailed modeling and forecasting of the power output of the power plant. Comparative experiments validated the model's prediction accuracy under different weather conditions, demonstrating the model's effectiveness and reliability in practical applications.

Fig. 3. Huadian Hubei Suizhou Suixian Yindian PV power plant.

The main steps of the experiment are shown in Fig. 4.

(1) Data acquisition: According to the central latitude and longitude of the experimental site, total radiation, diffuse radiation, and direct radiation data are obtained from FY-4 A data;

(2) Optical flow processing: The Farnebäck optical flow method is used to process the three types of radiation image data of size 256×256 extracted from FY-4 A data, obtaining the motion speed in the X and Y axes corresponding to the central point. Then, through certain mathematical calculations, the motion direction and speed of the three types of radiation for the central point and its eight neighboring areas are obtained. In this paper, the motion direction and speed data of the total radiation category are selected as input factors for subsequent multimodal data fusion;

(3) Data fusion: The time resolution of FY-4 A data is 60 min or 45 min, while the time resolution of numerical forecasts is 15 min. This paper uses linear interpolation to process FY-4 A data, generating multiple sets of data to unify its time resolution with that of numerical forecasts. Then, observed data, FY-4 A radiation data, radiation change direction and speed data, numerical forecast data, and numerical forecast features such as year, month, day, hour, and minute are fused to generate the final multimodal input factors;

(4) Power prediction: The CS-LSTM algorithm is used to predict the ultra-short-term power of a PV power station in Central China.

It is worth noting that in this study, the high temporal and spatial resolution of FY-4 A satellite data is crucial for capturing solar radiation variations caused by cloud cover. The performance of the model largely depends on the availability and accuracy of FY-4 A satellite data. If the quality of FY-4 A data is compromised, such as by missing data, noise, or inaccuracies, it will significantly impact the model's prediction accuracy. To mitigate the effects of such potential issues on model performance, future work could consider using data completion techniques such as cubic spline interpolation, Transformer-based multivariate time series interpolation, and Generative Adversarial Network (GAN)-based multivariate time series data imputation methods.

4.2. Quantitative evaluation experiment

This paper compares the predicted data with the actual PV power and uses prediction accuracy to quantitatively evaluate the effectiveness of the PV ultra-short-term power prediction model proposed in this paper. Due to the high variability in solar irradiance across different seasons (Iheanetu, 2022), from spring to winter, PV power exhibits significant fluctuations due to meteorological conditions such as solar radiation. Therefore, this paper selects data from 12 months and conducts experiments for spring, summer, autumn, and winter to comprehensively evaluate the performance of the proposed PV power ultra-short-term prediction model.

Table 1
Prediction comparison (Spring).

Model	March	Delta	April	Delta	May	Delta
CS-LSTM(O)	90.2571	0.8891	87.9501	0.4462	90.6394	0.1332
CS-LSTM(F)	91.1462		88.3963		90.7726	
LSTM(O)	91.0780	0.1117	88.2444	0.1205	90.4961	0.4267
LSTM(F)	91.1897		88.3649		90.9228	
AE-LSTM(O)	89.8207	1.0548	87.7061	1.7151	89.8849	1.4421
AE-LSTM(F)	90.8755		89.4212		91.3270	
GRU(O)	90.4156	0.6521	87.6940	0.1456	90.5578	0.5905
GRU(F)	91.0677		87.8396		91.1483	

Table 2
Prediction comparison (Summer).

Model	June	Delta	July	Delta	August	Delta
CS-LSTM(O)	89.9023	0.5421	89.4864	1.5747	88.4535	0.5603
CS-LSTM(F)	90.4444		91.0611		89.0138	
LSTM(O)	89.7552	0.5095	89.5694	0.2580	87.6522	0.5385
LSTM(F)	90.2647		89.8274		88.1907	
AE-LSTM(O)	89.9130	0.0898	90.0465	1.0981	87.0644	1.5324
AE-LSTM(F)	90.0028		91.1446		88.5968	
GRU(O)	87.5620	0.3024	89.5493	1.4108	88.1693	1.1315
GRU(F)	87.8644		90.9601		89.3008	

Table 3
Prediction comparison (Autumn).

Model	September	Delta	October	Delta	November	Delta
CS-LSTM(O)	88.2221	2.1936	90.4718	0.7247	90.5351	0.9376
CS-LSTM(F)	90.4157		91.1965		91.4727	
LSTM(O)	88.8077	0.8022	91.0510	0.5870	91.3823	0.0003
LSTM(F)	89.6099		91.6380		91.3826	
AE-LSTM(O)	87.9476	0.0445	90.2924	0.9057	90.2064	0.1845
AE-LSTM(F)	87.9921		91.1981		90.3909	
GRU(O)	89.4747	0.6429	91.3598	0.4309	91.4073	0.0832
GRU(F)	90.1176		91.7907		91.4905	

Table 4
Prediction comparison (Winter).

Model	December	Delta	January	Delta	February	Delta
CS-LSTM(O)	87.9278	0.4054	91.4741	0.9411	90.0238	1.4039
CS-LSTM(F)	88.3332		92.4152		91.4277	
LSTM(O)	87.9599	0.9445	89.9646	0.2261	91.5116	0.1072
LSTM(F)	88.9044		90.1907		91.6188	
AE-LSTM(O)	88.7030	0.1806	89.8840	0.2924	89.9047	1.2367
AE-LSTM(F)	88.8836		90.1764		91.1414	
GRU(O)	88.0897	0.4487	91.2522	0.4041	91.2998	0.1623
GRU(F)	88.5384		91.6563		91.4621	

To verify the effectiveness and applicability of the proposed method, this paper uses four algorithms, namely CS-LSTM, GRU, LSTM, and AE-LSTM, for comparative analysis. CS-LSTM(O), LSTM(O), AE-LSTM(O), and AE-LSTM(O) represent the control group without adding the optical flow input factor, while CS-LSTM(F), LSTM(F), AE-LSTM(F), and GRU(F) represent the experimental group with the optical flow input factor added. The results are shown in Table 1,Table 2, Table 3, and Table 4.

Tables 1–4 show the prediction results of the four algorithms under two scenarios: with and without the addition of optical flow features. Across the 12 months, all four algorithms have high prediction accuracy, and the results with added optical flow features are consistently stronger than those without, indicating the general applicability of the framework proposed in this paper. Notably, due to the variable weather in summer, the prediction accuracy sees a significant improvement, while the improvement is limited in autumn when the weather is mostly sunny. It is worth mentioning that the AE-LSTM algorithm,

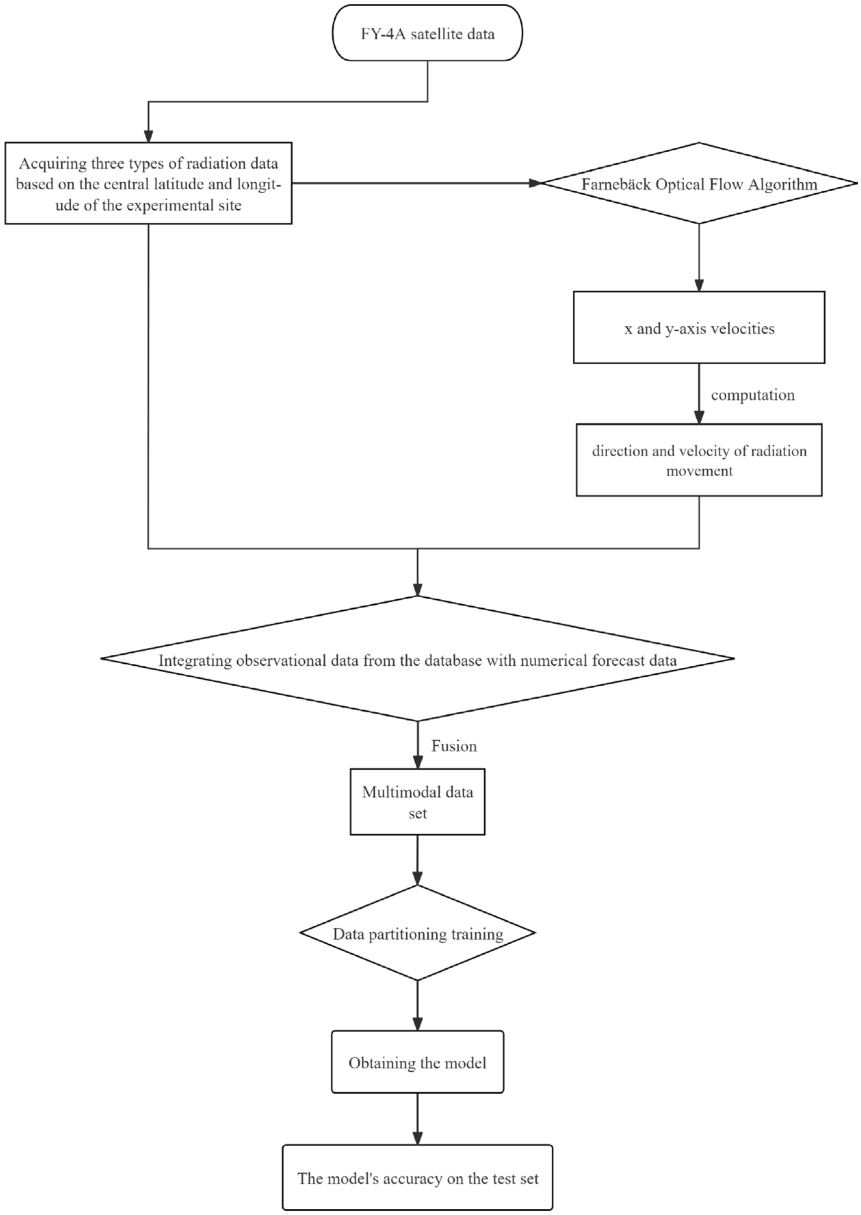

Fig. 4. Experimental flowchart.

which combines the feature extraction capability of the autoencoder with the sequence modeling ability of LSTM, shows better performance than the single LSTM in our experiments. The main algorithm used in this paper, CS-LSTM, combines the local feature extraction capability of CNN with the ability of LSTM to handle long-term dependencies, and exhibits stronger robustness and generalization ability compared to the other three algorithms in the experiments.

To further validate the effectiveness of the simulation results, we selected real measurement data from March, June, September, and December and compared them with the predicted data from both the experimental group and the control group. The specific data are shown in Fig. 5, Fig. 6, Fig. 7, and Fig. 8. These charts illustrate the trends in real measurement data, experimental group data, and control group data for each month, where "Real" represents the real measurement data, "OF" represents the experimental group with added optical flow input features, and "Original" represents the control group without the added optical flow input features. It is worth noting that the real measurement data for June contained significant anomalies over two consecutive days, which were removed during data preprocessing. As a result, the time span on the horizontal axis in Fig. 6 is slightly shorter than in the other three figures.

As shown in Figs. 5 to 8, the experimental group with added optical flow features consistently outperforms the control group without these features in terms of prediction accuracy, which aligns with the results presented in the earlier tables. Particularly in months with significant weather variations, the experimental group's predictions exhibit a higher degree of fit with the actual measurement data, validating the robustness and applicability of the proposed model in handling complex meteorological conditions. This further demonstrates the effectiveness and reliability of the Multidimensional Feature Fusion and Error Progressive Approximation (MDFF-EPA) prediction algorithm.

4.3. Four weather types experiment

As the prediction model faces different challenges under varying weather conditions, forecasting the photovoltaic (PV) power output under all climatic conditions is a significant test of the comprehensive performance of the model proposed in this paper. This study categorizes

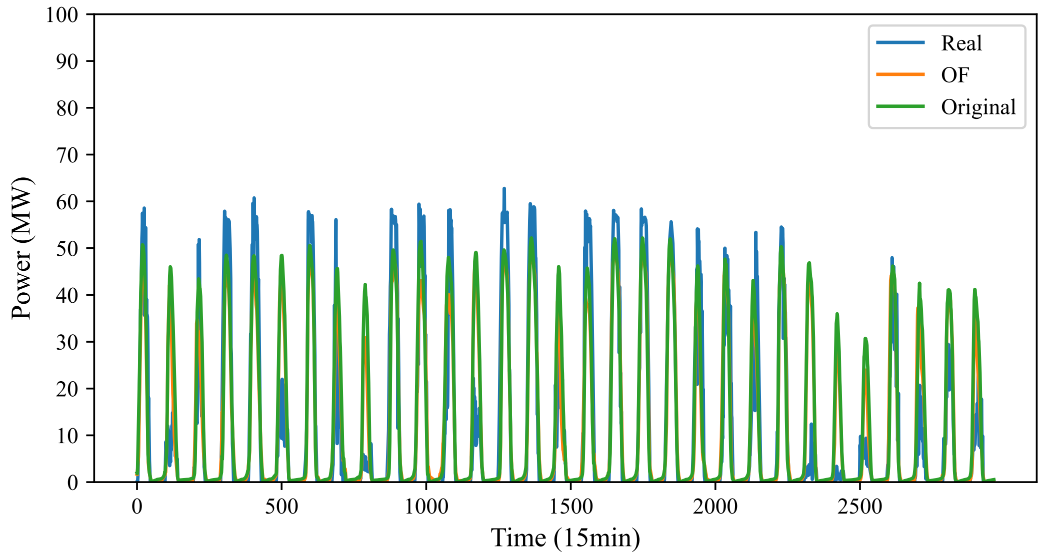

Fig. 5. Comparison of actual measurement data and predicted data for March (Based on CS-LSTM).

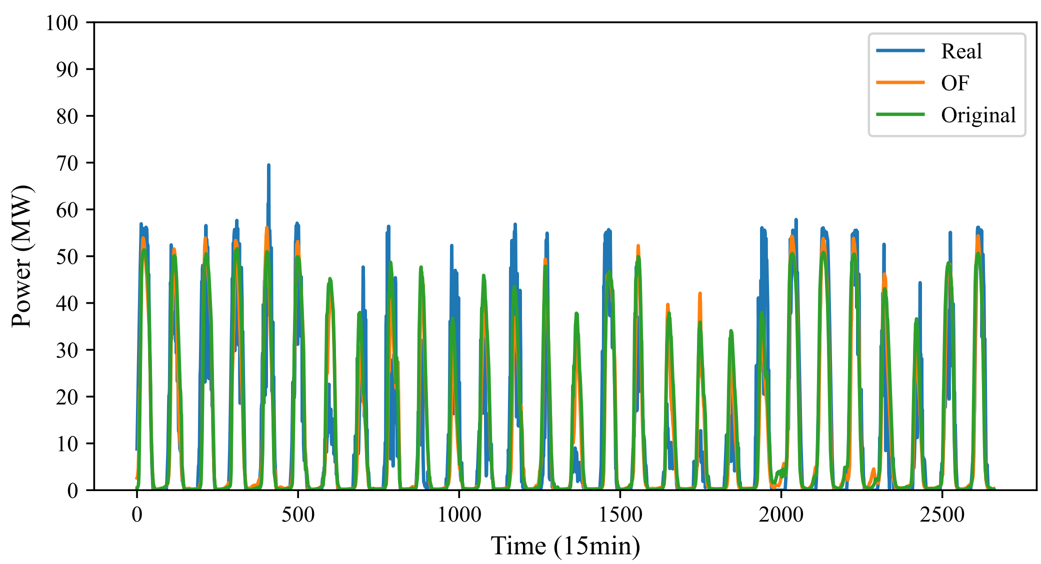

Fig. 6. Comparison of actual measurement data and predicted data for June (Based on CS-LSTM).

Table 5
Comparison of prediction accuracy for four weather types.

Weather type	Original	OF
Sunny	86.7611	91.2592
Cloudy	88.7559	92.4281
Rainy	89.5913	93.3674
Heavy rain	90.0492	92.0370

typical weather into four types: sunny, cloudy, rainy, and heavy rain. Then, based on observed radiation data, July 5 and July 2, 2019, July 16, 2019, and March 8, 2020, are selected as representatives of the sunny, cloudy, rainy, and heavy rain weather types, respectively. ‘Original’ denotes the control group without the addition of optical flow input factors, while ‘OF’ represents the experimental group with added optical flow input factors. The overall prediction accuracy of the aforementioned four days is shown in Table 5.

Table 5 shows that in all four weather types, the cases with added optical flow input factors outperform those without optical flow input factors. Since PV output is almost zero at night (Liang et al., 2023), to further verify the performance of the proposed PV power ultra-short-term prediction model under different meteorological conditions in detail, this paper, based on the CS-LSTM algorithm, selects data from the 8:00–18:00 period of the aforementioned four days for a comparison of prediction accuracy. The predicted power without optical flow input factors (Original), with optical flow input factors (OF), and the actual PV power output of the PV station (Real) are compared, as shown inFig. 9 and Fig. 10.

Fig. 9 and Fig. 10 visually demonstrate that under different meteorological conditions, the experimental group with added optical flow input factors consistently outperforms the control group, indicating that the model proposed in this paper has good robustness and strong overall predictive performance. Similarly, to verify the comprehensive performance of the proposed model from another dimension, this paper continues to select data from 8:00-18:00 of the aforementioned four days to calculate the RMSE.

As shown in Fig. 11, for the four weather types of sunny, cloudy, rainy, and heavy rain, the RMSE of the experimental group is smaller than that of the control group without added optical flow input factors. Therefore, the model proposed in this paper can accurately predict the

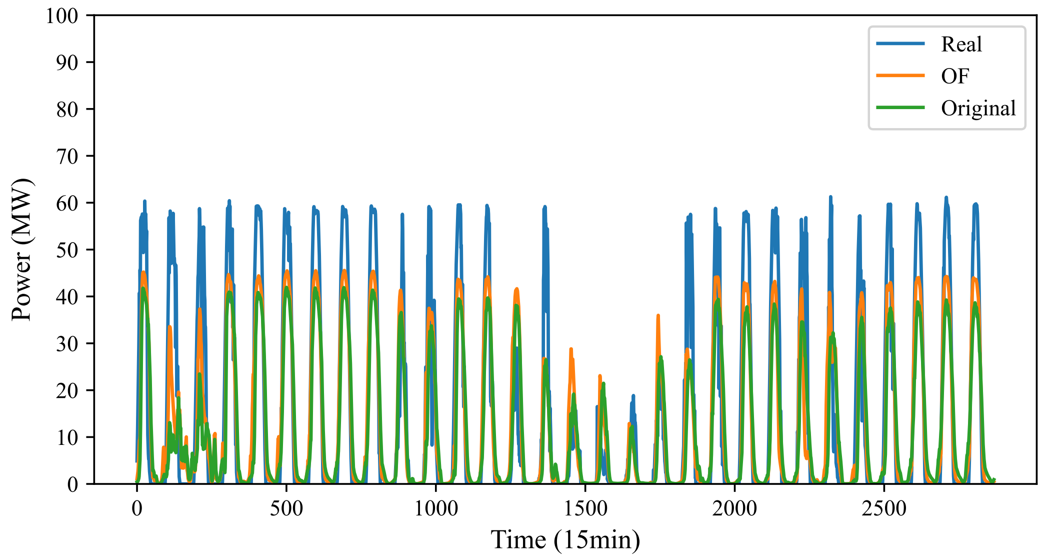

Fig. 7. Comparison of actual measurement data and predicted data for September (Based on CS-LSTM).

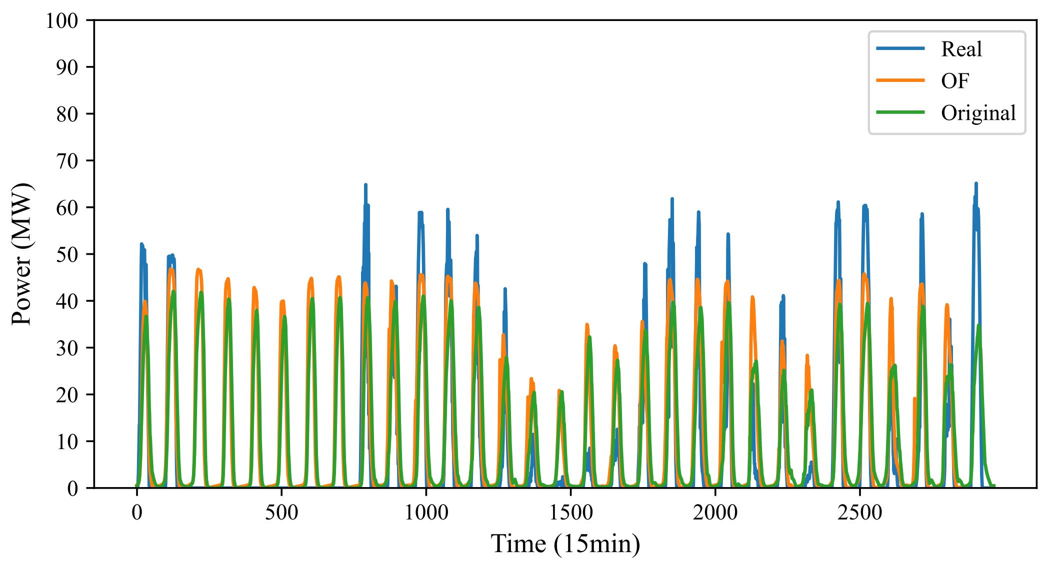

Fig. 8. Comparison of actual measurement data and predicted data for December (Based on CS-LSTM).

ultra-short-term power of PV under different meteorological conditions, and it also has strong robustness.

In summary, this study proposes an ultra-short-term PV power forecasting model based on FY-4 A satellite data, utilizing a Multidimensional Feature Fusion and Error Progressive Approximation (MDFF-EPA) algorithm to enhance prediction accuracy. All four algorithms (CS-LSTM, GRU, LSTM, and AE-LSTM) demonstrated higher prediction accuracy after incorporating optical flow features, with the CS-LSTM algorithm showing stable and significant improvements across multiple seasons.

Compared to existing research, the innovation of this study lies in introducing optical flow features as model inputs, enabling more accurate capture of the impact of cloud movement on solar radiation. Although commonly used methods in other studies, such as ARIMA, ANN, and SVM models, can also improve prediction accuracy, they often involve complex model structures and high computational costs. The proposed model in this study maintains lower complexity while enhancing prediction performance, especially under variable summer weather conditions. This aligns with results in existing literature but offers greater practical applicability and innovation.

Experimental results indicate that incorporating optical flow features improves the accuracy of PV power forecasts, particularly under conditions of significant weather changes such as cloudy and rainy days. The CS-LSTM algorithm, combining the advantages of Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, is better equipped to handle spatial and temporal information, further enhancing the robustness and generalization capabilities of the model.

The advantage of this study lies in proposing a method that combines optical flow features with multidimensional feature fusion, validating its applicability and high prediction accuracy across different seasons and weather conditions. However, there is still room for improvement in terms of the model's portability across datasets from different provinces, handling missing and noisy satellite data, and the

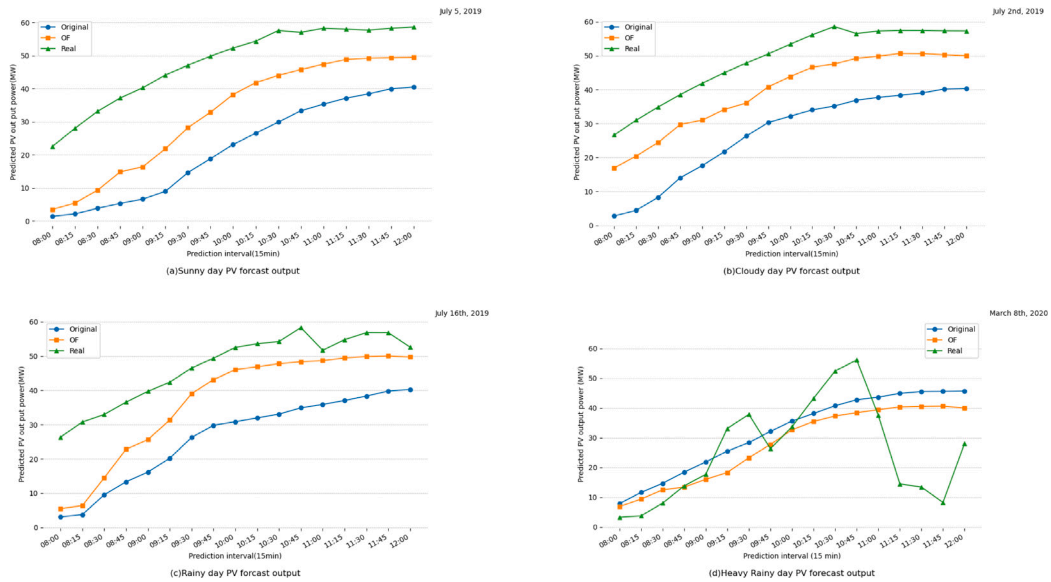

Fig. 9. Validation results of prediction data in different meteorological conditions for the morning (8:00–12:00).

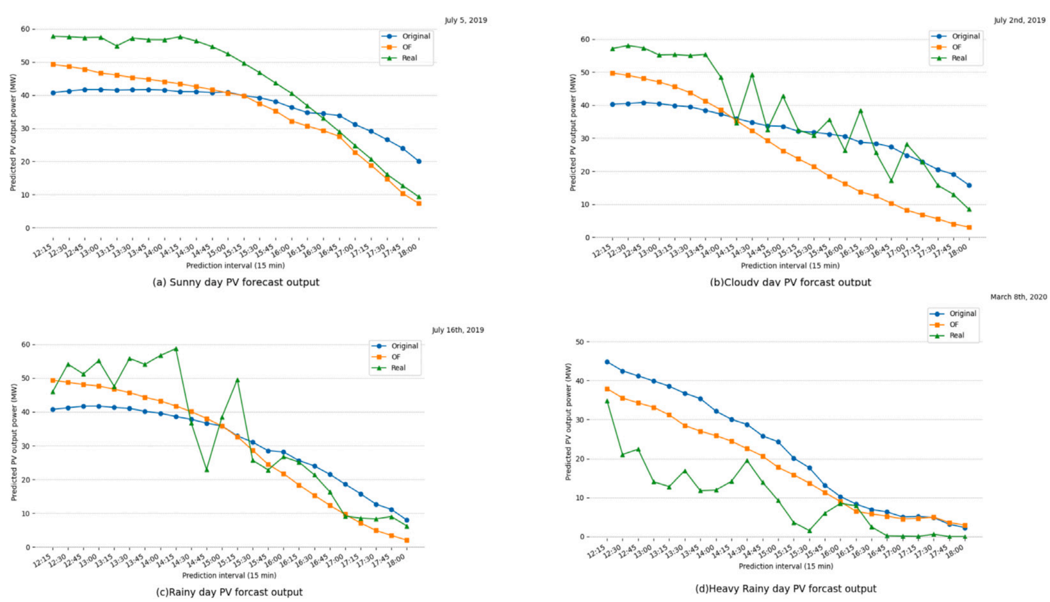

Fig. 10. Validation results of prediction data in different meteorological conditions for the afternoon (12:15–18:00).

suitability of the optical flow algorithm for dealing with large displacements and textureless areas. Additionally, higher frequency satellite data (e.g., 15-minute intervals) would further enhance prediction accuracy.

5. Conclusion and future work

This paper successfully proposes an ultra-short-term PV power forecasting model based on FY-4 A satellite data, utilizing a Multidimensional Feature Fusion and Error Progressive Approximation (MDFF-EPA) algorithm to achieve efficient ultra-short-term PV power forecasting. The main contributions of this paper are as follows:

In terms of theoretical contributions, compared to the complex hybrid forecasting models currently prevalent in the field of ultra-short-term PV power forecasting, the proposed model reduces structural complexity while enhancing prediction performance to a certain extent. This innovative approach provides a simple and efficient solution for ultra-short-term PV power forecasting, expanding the theoretical foundation of the field.

Experimental results indicate that the addition of optical flow features led to a certain improvement in the accuracy of all four forecasting algorithms. The improvement is most notable under variable summer weather conditions. Specifically, the experimental group's prediction accuracy exceeded the control group by over 1% in certain months. Under clear, cloudy, rainy, and heavy rain conditions, the RMSE percentages decreased by 15.04%, 11.97%, 14.84%, and 18.86%, respectively.

Considering the impact of cloud movement on the ultra-short-term power generation of PV plants, this paper incorporates optical flow radiation input factors into the model and validates it using actual

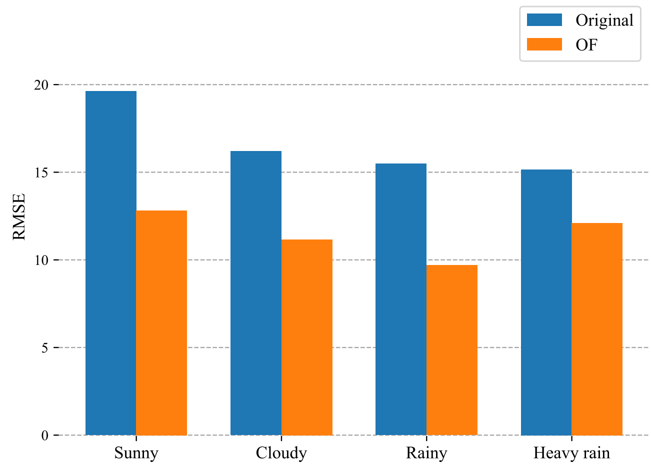

Fig. 11. RMSE evaluation index under different meteorological conditions (Based on CS-LSTM).

output data from a PV power plant in central China, demonstrating the model's practical engineering value.

The proposed model exhibits broad applicability and strong robustness, but it also has certain limitations. For instance, the model's portability across datasets from different provinces is insufficient, and there are issues with missing and noisy satellite data. The applicability of the optical flow algorithm is particularly weak when dealing with large displacements and textureless areas. Moreover, higher temporal resolution data (such as 15-minute interval satellite cloud images) would further improve prediction accuracy.

This study contributes a new efficient forecasting model to the field of ultra-short-term PV power forecasting, significantly enhancing prediction accuracy and model practicality by combining optical flow features and multidimensional feature fusion. This provides robust theoretical support and practical references for future research and applications in PV power forecasting, driving further development in the field.

Future research can improve the model's portability to adapt to different regional characteristics, optimize the quality and temporal resolution of satellite data, and conduct more comprehensive experiments by introducing more dimensions of data and more complex meteorological conditions to further validate and enhance the model's robustness and accuracy.

CRediT authorship contribution statement

Renfeng Liu: Writing – original draft, Validation, Software, Data curation, Conceptualization. **Zhuo Min:** Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation. **Desheng Wang:** Formal analysis, Data curation, Conceptualization. **Yinbo Song:** Software, Formal analysis. **Chen Yuan:** Writing – review & editing, Supervision, Methodology. **Gai Liu:** Visualization, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used is confidential. For access to specific data, please contact the corresponding author.

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